QUANTUM ERROR CORRECTION (QEC) — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 29, 2026

Diagram-based questions on Quantum Error Correction (QEC)

Question 1

10.0 points

Analyze the circuit shown below of a stabilizer code. Identify how the syndrome measurement helps in error detection and explain the role of stabilizers in quantum error correction.

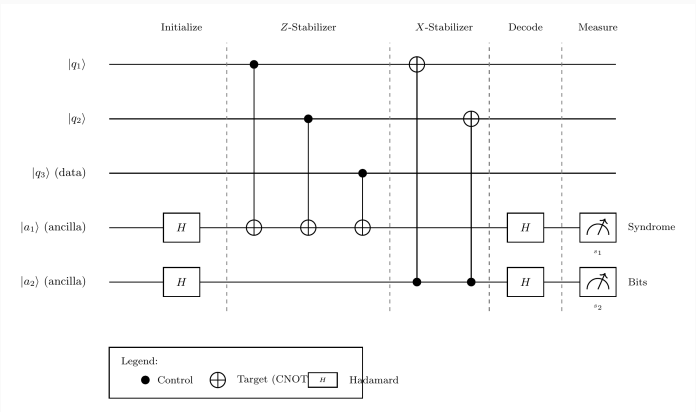

Fig. 1. Diagram for Question 1

Question 2

10.0 points

Examine the circuit shown below illustrating the 3-qubit bit-flip code with its corresponding error detection circuit. Explain the error correction process and identify the role of each circuit component in detecting a single bit-flip error.

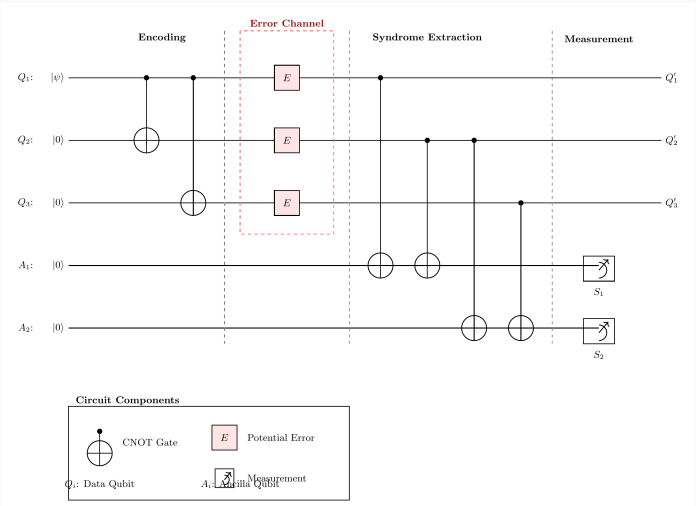

Fig. 2. Diagram for Question 2

Question 3

10.0 points

Examine the surface code lattice in the diagram below. Describe how logical qubits are encoded on the lattice and discuss how the layout is used for error correction.

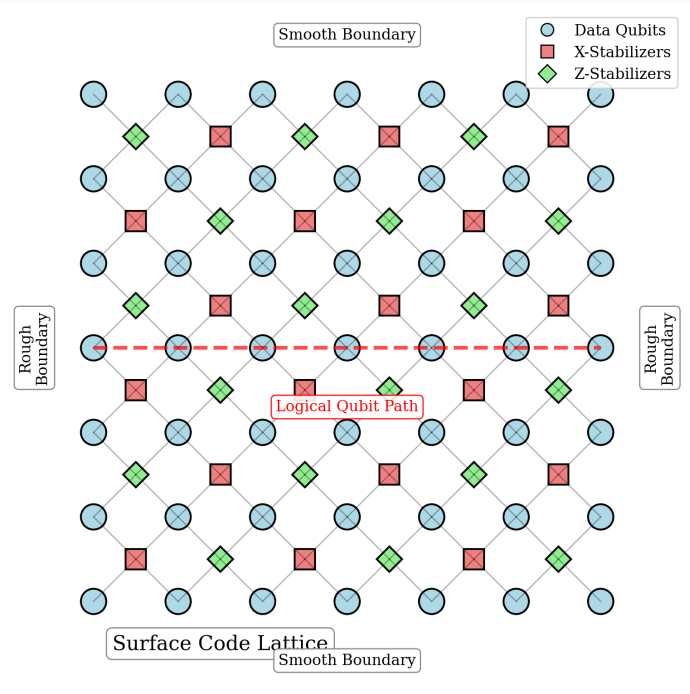

Fig. 3. Diagram for Question 3

Question 4

10.0 points

Examine the diagram below illustrating fault-tolerant gate teleportation. Discuss how teleportation is utilized to achieve fault-tolerance in quantum gates and what protections it offers against errors.

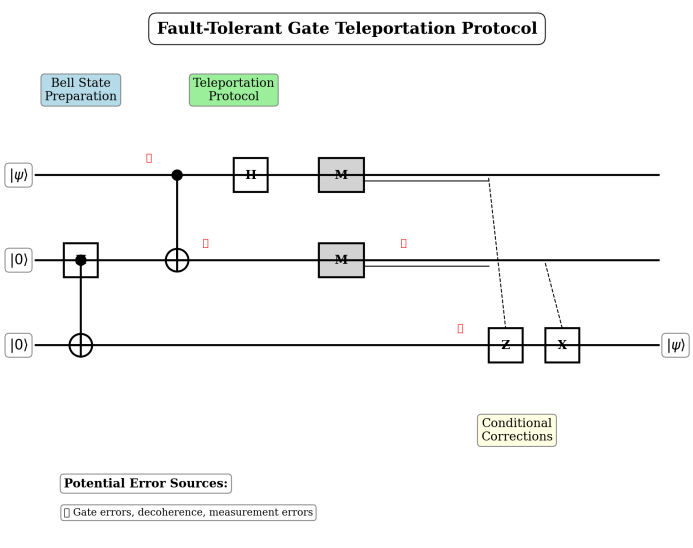

Fig. 4. Diagram for Question 4

Question 5 10.0 points

Interpret the threshold theorem in the diagram below, which illustrates error rates in quantum error correction. Explain how the error rate threshold is defined and its significance in achieving fault-tolerant quantum computation.

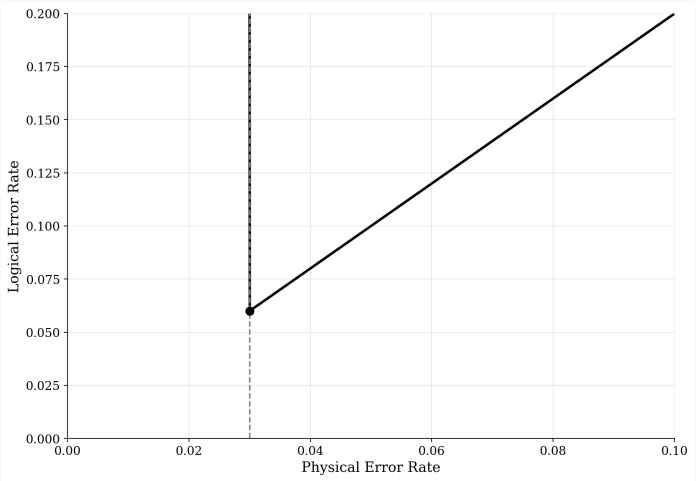

Fig. 5. Diagram for Question 5